

PROJECT

Geospatial Statistics (GNR640)

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Geospatial Interpolation of Climate Variables over CONUS using USCRN and ERA5

1. Objective

The objective of this study is to generate daily gridded climate datasets at a $2^\circ \times 2^\circ$ resolution over the contiguous United States using station-based observations. Multiple spatial interpolation techniques are evaluated and validated using ERA5 reanalysis data.

2. Data Description

The analysis uses daily observations from the United States Climate Reference Network (USCRN) for the period 2006–2021.

A total of 208 stations across CONUS were used after filtering.

The variables considered include:

- Air temperature
- Precipitation
- Relative humidity
- Soil moisture (10 cm depth)
- Soil temperature

Additionally, ERA5 reanalysis data for the year 2015 was used for validation.

3. Methodology

A uniform grid of $2^\circ \times 2^\circ$ resolution (420 grid cells) was created over CONUS.

The following interpolation methods were applied:

- Inverse Distance Weighting (IDW)
- Linear interpolation
- Radial Basis Function (RBF) interpolation
- Kriging (spherical, circular, and gaussian variograms)
- Model Evaluation

Model performance was evaluated using:

- Root Mean Square Error (RMSE)
- Correlation coefficient
- Bias

A 5-fold cross-validation approach was used.

4. ERA5 Comparison

The interpolated grids were compared with ERA5 reanalysis data. (ERA5 data was regridded to match the 2° grid)

Comparison metrics: RMSE & Spatial correlation

5. Results

Best Model for every variable with RMSE and correlation:

variable	model	rmse	correlation
Air temperature	RBF linear	2.288804792	0.958107185
Precipitation	Linear	5.26530469	0.6431629
Relative humidity	RBF linear	9.917792674	0.792481209
Soil moisture 10cm	Kriging spherical	0.096967484	0.583067242
Soil temperature	RBF linear	2.72287601	0.948617901

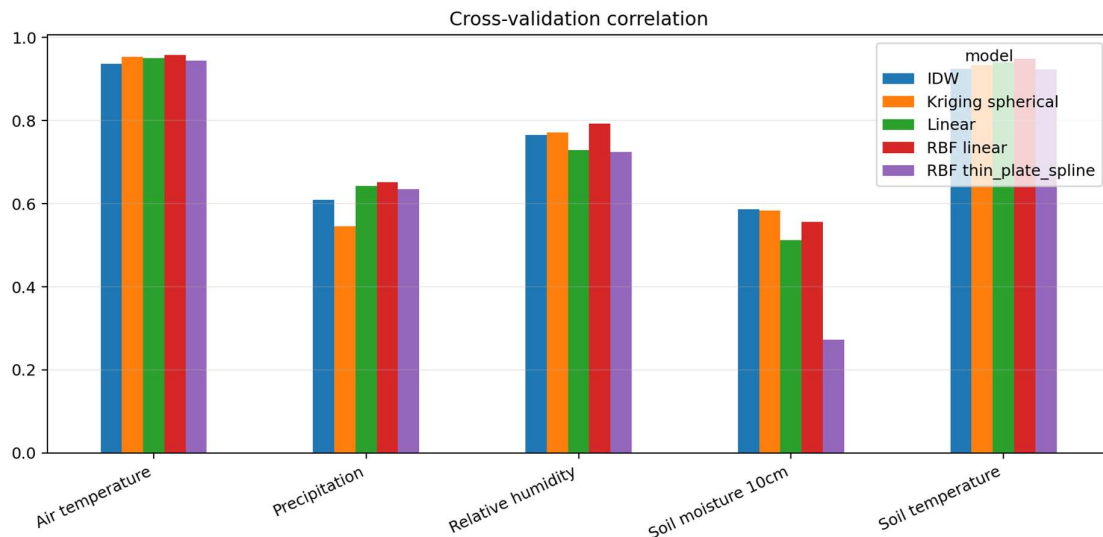
The results indicate that RBF linear performs best for most variables, particularly for air temperature and soil temperature, showing high correlation (>0.94) and low RMSE.

Precipitation shows comparatively lower correlation due to its high spatial variability and sparsity.

Soil moisture exhibits moderate performance, with kriging spherical providing the best results among tested methods.

The cross-validation results indicate that:

- IDW and Kriging consistently perform best across variables
- RBF methods provide moderate performance
- Linear interpolation shows instability in regions with sparse station coverage



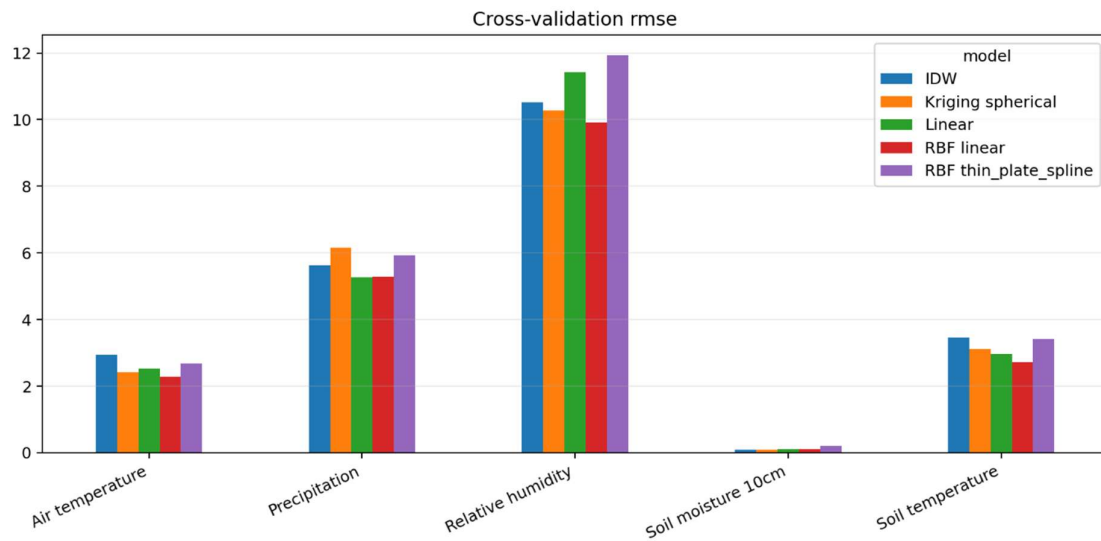
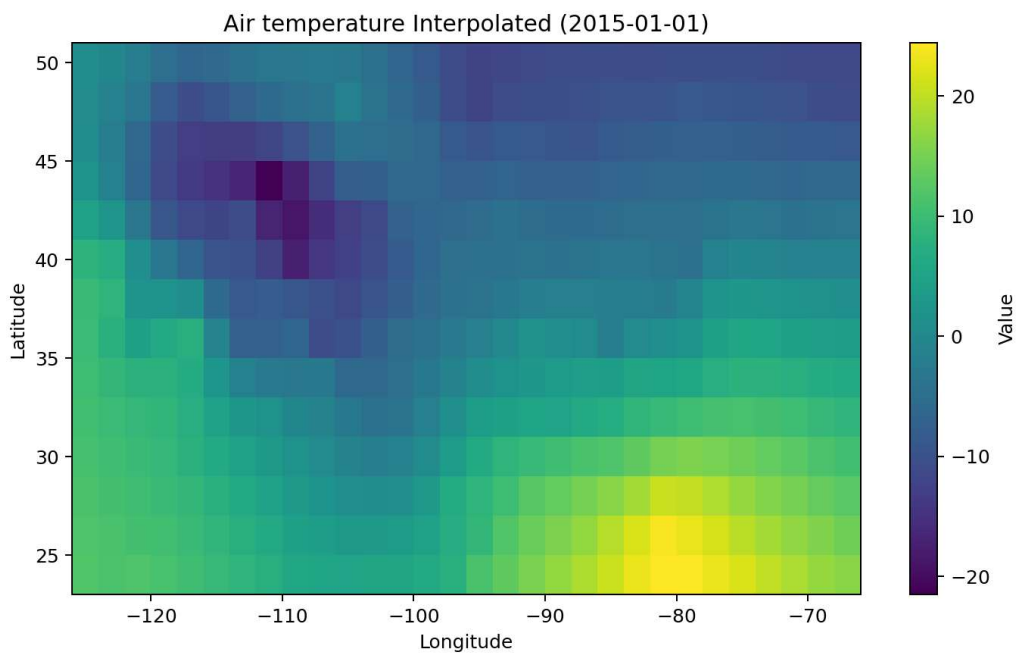


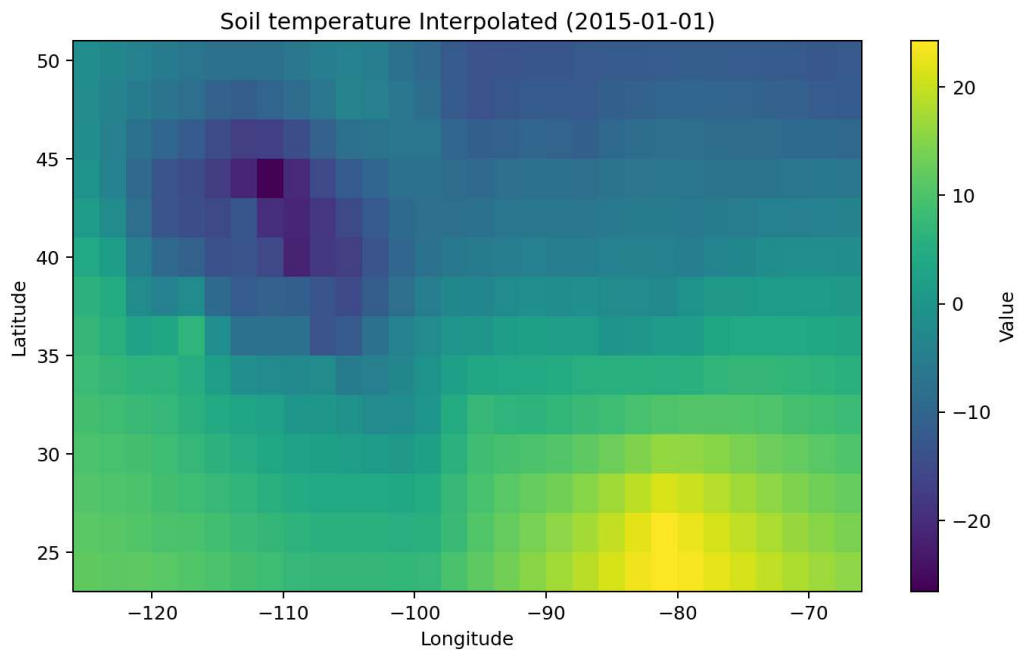
Figure 1–2: Cross-validation RMSE and correlation for different interpolation methods

The generated spatial maps show:

- Smooth gradients for air temperature and soil temperature
- High variability in precipitation fields
- Consistent spatial structures across interpolation methods

Kriging produces smoother surfaces, while IDW preserves local variability.





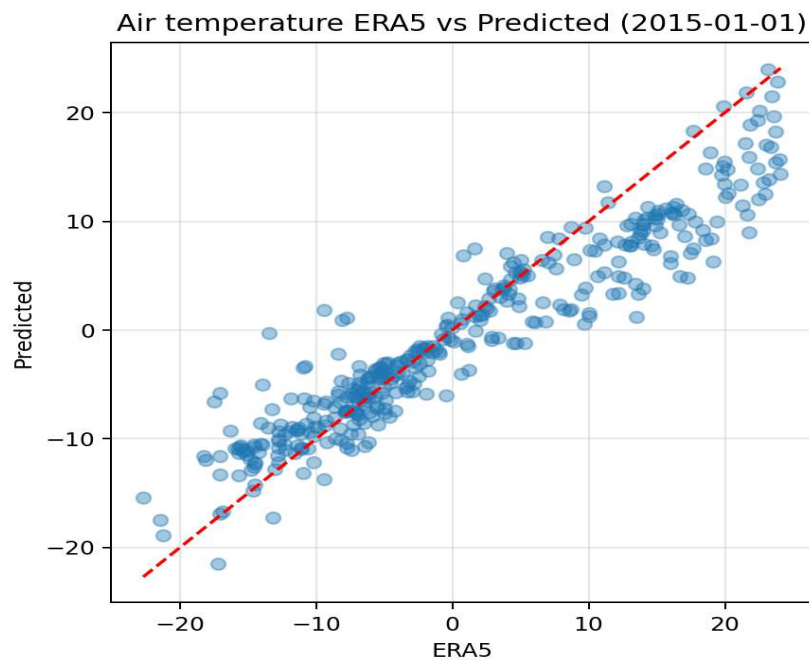
Example spatial distribution of interpolated climate variable over CONUS

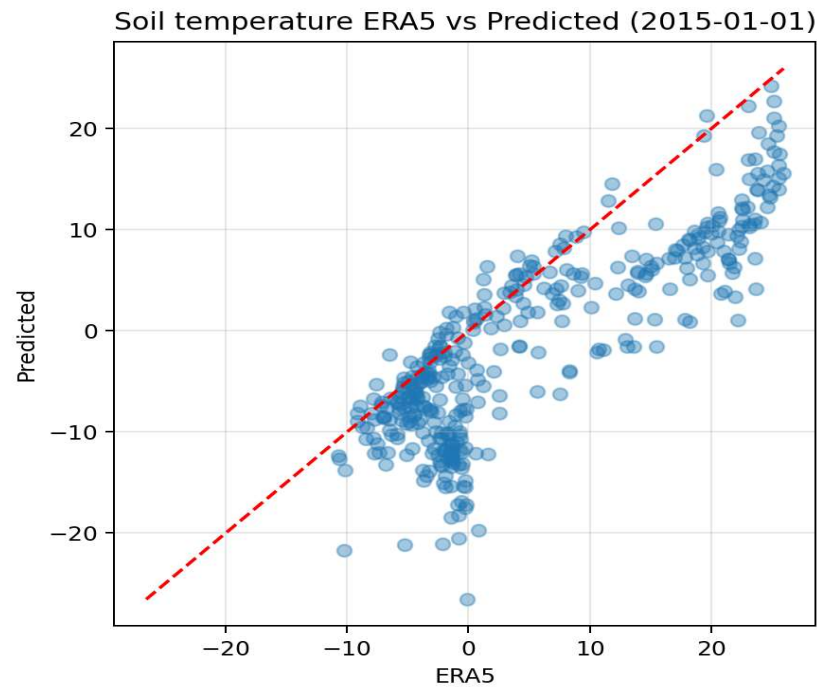
ERA5 Comparison

Comparison with ERA5 shows:

- Strong agreement for air temperature
- Moderate agreement for relative humidity and soil variables
- Higher error for precipitation due to its spatial variability

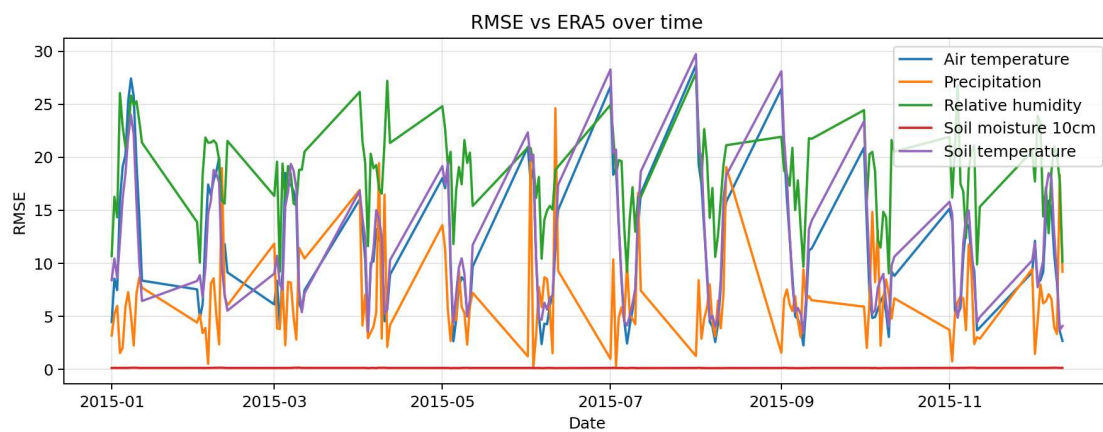
Scatter plots show clustering around the 1:1 line, indicating good agreement.





Temporal Performance

RMSE varies across time but remains stable overall, indicating consistent model performance.



RMSE variation over time for different variables

Statistical Analysis

Statistical summaries indicate:

- Stable mean and variance across variables
- Seasonal cycles consistent with climatological expectations
- Peak temperatures during summer months

- Higher variability in precipitation

Kolmogorov–Smirnov (KS) tests show reasonable similarity between observed and interpolated distributions.

6. Conclusion

This study demonstrates that:

- IDW and Kriging are the most reliable interpolation methods
- Interpolated fields closely match ERA5 data, especially for temperature